[Riya Patil]

[CSE-DS]

Company: MasterCard

Job Profile: Software Engineer I

Offer Type: [Placement Only]

Internship Stipend: NA

CTC: 13.40 LPA (Fixed: 11 LPA + Joining Bonus: 0.50 L + Variable: 1.90 LPA)

Location: Pune

Process Date: PPT(16 Sept, 4.30 PM - Online), OA(16 Sept, 9 PM - Online), Interview(19 Sept, 10 AM onwards - In Office)

Placement Process:

1. OA - It consisted of 2 Coding Questions
2. 3 Interview Rounds - 2 Technical + 1 HR

1. **[Online Assessment]** [35 candidates were shortlisted based on Resume]

Platform & Duration:
The first round was an **online proctored assessment** conducted on **AssessHub**. The test was scheduled at 9:00 PM and conducted from home with a total duration of **1 hour**.
Note:
The platform was very strict regarding malpractice. Even attempting to copy–paste code (Ctrl+C / Ctrl+V) triggered alerts. Hence, all code had to be typed manually.

Test Pattern:
The assessment consisted of **2 coding questions**, both of which were logically easy but required clarity of concepts.

1. Encryption & Decryption Program
   - Based on ASCII values and bit manipulation
   - Required understanding of character encoding and logical operations.
2. Pattern-Based Question
   - Print the mirror image and water image of a given pattern

Shortlisting:
Around 35 students appeared for this round, 9 students were shortlisted for the interview process.

2. **[Technical Interview – 1]** [Duration: ~ 40–50 minutes]

Interview Experience:

The interview was conducted by a **Platform Engineer (Team Lead)** and focused on **core fundamentals, problem-solving approach, and mindset**.

Topics & Questions Asked:

1. Introduction
   - Asked to briefly introduce myself.
2. Core Java Concepts
   - Difference between **JVM, JDK, and JRE**
   - Why Java is called **"Write Once, Run Everywhere (WORE)"**
3. Project Discussion
   - Asked to explain **how I develop a project from scratch**
   - I explained my **major project**, covering:
     - Problem statement analysis
     - Researching possible solutions
     - Evaluating solutions based on **feasibility, viability, and business impact**
     - Selecting the tech stack
     - Task division, development, and unit testing
4. Conceptual Question
   - Explained **Machine Learning** as if explaining it to a **5-year-old** (tested clarity of understanding).
5. Coding & Logical Questions
   - Reverse a string (I clarified whether it should be **in-place**)
   - Complete a **number series**
   - Write **pseudo-code for a pattern**
6. Scenario-Based Question
   - Asked whether I would work on a **simple or less popular technology** even if it may not be useful in the long term
   - This question tested **attitude and professionalism**, and required a **balanced, diplomatic answer**

Key Takeaways & Tips:

- Always **clarify problem statements and edge cases**
- Focus on **approach**, not just the final answer
- Be honest but diplomatic in scenario-based questions
- Communicate clearly and confidently

3. **[Technical Interview – 2]** [Duration: ~ 1 hour 15 minutes]

Interview Experience:
This round was conducted by a **Senior Developer** and was **deeply technical**. The interviewer explicitly mentioned that he would go through **each item mentioned in my resume**.

Topics & Questions Asked:

1. **Object-Oriented Design**
   - Given an **employee scenario**, asked to:
     - Design class hierarchy
     - Use appropriate data structures
     - Think from a **production-level perspective**
2. **Logical & Coding Questions**
   - Add, fetch, and delete employees based on conditions
   - Example:
     - Find employee with **highest salary**
     - If salaries are same, return the employee who **joined most recently**
3. **DBMS**
   - Designed an **ER diagram** for the employee system
   - Wrote **4–5 SQL queries**
   - Differences between **SQL and NoSQL**
   - Given scenarios and asked which database to use and **why**
4. **Data Structures**
   - Code to find the **meeting point of two linked lists**
5. **Internship Discussion (Kafka)**
   - Why Kafka instead of simple APIs
   - How Kafka improved **efficiency and scalability**
   - Where Kafka was used in the project
6. **Project-Based Questions**
   - Detailed discussion on projects mentioned in my resume

Key Takeaways & Tips:

- Know **everything written in your resume in depth**
- Practice **OOP design questions**
- Revise **DBMS, SQL queries, and linked list problems**
- Be prepared to justify **technology choices**

4. **[HR Interview]** [Duration: ~ 20–25 minutes]

Interview Experience:
This was a conversational round focused on **personality, awareness, and fit**.

Questions Asked:

- Feedback on my previous two interviews
- Names of interviewers (to check **attentiveness**)
- Internship experience (**internships are always a plus**)
- Family background
- Willingness to **relocate to Pune**

Overall Feel:
The HR round was friendly and more about understanding me as a person rather than testing technical skills.

[2 students were offered the job]

Useful Tips:

- **Striver's DSA Sheet** is honestly one of the **best resources** to practice DSA. If you're confused about where to start, just follow the sheet consistently.
- **Keep your fundamentals very clear.** No matter how much you code, interviewers always test your basics.
- Just like many others, **I initially thought placement preparation means only focusing on DSA**. But that's not true.
  You should prepare **SQL queries** with the same seriousness as DSA coding.
- For **SQL practice**, **SQL50** is a great place to strengthen your basics.
  Additionally, I used **ChatGPT** to generate **scenario-based SQL questions and complex queries**, which helped me think from an interview perspective.
- For **Aptitude preparation**, I found **CareerRide** to be a really good and reliable resource.
- Make sure you revise **DBMS and OS fundamentals**, as they are frequently asked in interviews.
- And most importantly, **communication skills**.
  Even if you know the answer, you should be able to clearly explain your approach and thought process. Interviewers pay a lot of attention to this.